# INFOSYS SE — MATCHSTICK, CUBE & SPATIAL PUZZLES

**30 Rigorous Spatial Questions • Minimal Matchstick Removals, 7×7×7 Cube Paint Cuts & Triangle Counting**

## SECTION 1: Matchstick, Cube & Spatial Logic Puzzles (30 Questions)

**[College ITM Mock Paper 2]**

***Directions:*** *Matchstick Grid Puzzle: A row of 5 unit squares is built using 16 matchsticks.*

**Q1. What is the minimum number of matchsticks to remove so that exactly 3 complete unit squares remain?**

a) 1
b) 2
c) 3
d) 4

**Answer: A**

**Logic / Trick:** The matchstick shared between the 2nd and 3rd unit squares is a side of both. Removing that single internal stick breaks both squares at once, leaving the 1st, 4th, and 5th squares intact (exactly 3 complete squares) with only 1 stick removed.

**[College ITM Mock Paper 3]**

***Directions:*** *Matchstick L-Shape Puzzle: A figure is made of 3 unit squares arranged in an L-shape using 10 matchsticks in total.*

**Q2. What is the minimum number of sticks to remove so that no complete square remains?**

a) 1
b) 2
c) 3
d) 4

**Answer: B**

**Logic / Trick:** The stick shared between the top-left square and top-right square is a side of both. Removing it breaks both squares in one move. Removing one more exclusive stick of the bottom square breaks the third. Minimum sticks = 2.

**[College ITM Mock Paper 5]**

***Directions:*** *Matchstick Plus-Sign Puzzle: A cross figure is made of 5 unit squares (1 central square + 4 outer squares attached) using 16 matchsticks.*

**Q3. What is the minimum number of sticks that must be removed so that no complete square remains?**

a) 3
b) 4
c) 5
d) 6

**Answer: B**

**Logic / Trick:** Each of the 4 outer squares shares exactly one stick with the centre square and no sticks with each other. Removing all 4 shared inner sticks breaks all 4 outer squares and simultaneously destroys the centre square as well. Minimum = 4.

**[College ITM Mock Paper 6]**

***Directions:*** *Matchstick 2x3 Grid: A 2 x 3 grid of unit squares is built from 17 matchsticks and contains 8 squares in total (six 1x1 squares and two 2x2 squares).*

**Q4. What is the minimum number of sticks that must be removed so that no square of any size remains?**

a) 3
b) 4
c) 5
d) 6

**Answer: B**

**Logic / Trick:** Exhaustive verification over all 17 sticks and 8 squares confirms that removing 3 sticks leaves at least one 1x1 or 2x2 square intact. Removing a specific set of 4 sticks destroys all six 1x1 and both 2x2 squares. Minimum = 4.

**[College ITM Mock Paper 6]**

***Directions:*** *Cube Painting & Cutting: A solid wooden cube of side 7 units is painted on all 6 faces and then cut into 1-unit smaller cubes (7 x 7 x 7 = 343 small cubes).*

**Q5. How many small cubes have at most 1 painted face?**

a) 275
b) 125
c) 150
d) 335

**Answer: A**

**Logic / Trick:** Total cubes = 7^3 = 343. "At most 1 painted face" = (0 painted faces) + (1 painted face). Cubes with 0 painted faces = (7-2)^3 = 5^3 = 125. Cubes with 1 painted face = 6 * (7-2)^2 = 6 * 25 = 150. Sum = 125 + 150 = 275.

**[College ITM Mock Paper 1]**

***Directions:*** *Geometry & Figure Counting: In triangle ABC, medians AD, BE and CF are drawn and meet at centroid G.*

**Q6. How many triangles are there in the figure in total?**

a) 12
b) 14
c) 16
d) 18

**Answer: C**

**Logic / Trick:** 6 small triangles meeting at centroid G. 3 triangles formed by pairing adjacent small triangles across medians (taking 2 at a time = 6). 3 large triangles divided by each median (taking 3 at a time = 3). Plus the whole triangle ABC = 1. Total = 6 + 6 + 3 + 1 = 16.

**[College ITM Mock Test 1 (Q51 - Matchstick)]**

***Directions:*** *Matchstick Geometry: A 2×2 grid of unit squares is made using 12 matchsticks (giving 4 small 1×1 squares and 1 large 2×2 square).*

**Q7. What is the minimum number of matchsticks to remove so that no square of any size remains?**

(a) 2
(b) 3
(c) 4
(d) 5

**Answer: (b) 3**

**Logic / Trick:** Each 1×1 square requires at least 1 stick removed. Removing the central horizontal stick breaks 2 squares; removing the central vertical stick breaks the other 2; removing 1 boundary stick breaks the 2×2 square. Exactly 3 sticks break all 5 squares. Computer proof verifies minimum = 3.

**[College ITM Mock Test 1 (Q52 - Circular Wheel)]**

***Directions:*** *Circular Wheel Logic: Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.*

**Q8. Wheel 1: T=3, R=5, B=4, L=7 -> Centre=14. Wheel 2: T=2, R=6, B=5, L=3 -> Centre=7. Find Centre for Wheel 3: T=4, R=3, B=6, L=9.**

(a) 21
(b) 27
(c) 30
(d) 33

**Answer: (c) 30**

**Logic / Trick:** Rule: Centre = (Top × Bottom) + (Left - Right). Wheel 1: (3×4) + (7-5) = 12 + 2 = 14. Wheel 2: (2×5) + (3-6) = 10 - 3 = 7. Wheel 3: (4×6) + (9-3) = 24 + 6 = 30.

**[College ITM Mock Test 1 (Q53 - Matrix Logic)]**

***Directions:*** *3×3 Numerical Matrix: Row 1: [3, 4, 25]; Row 2: [5, 12, 169]; Row 3: [6, 8, ?].*

**Q9. Which number replaces the question mark in the 3×3 matrix?**

(a) 64
(b) 100
(c) 196
(d) 144

**Answer: (b) 100**

**Logic / Trick:** Rule: Third column = (First)² + (Second)² (Pythagorean Triples). Row 1: $3^2 + 4^2 = 9 + 16 = 25$. Row 2: $5^2 + 12^2 = 25 + 144 = 169$. Row 3: $6^2 + 8^2 = 36 + 64 = 100$.

**[College ITM Mock Test 1 (Q54 - Cubic Pattern)]**

***Directions:*** *Mathematical Sequence: 2, 12, 36, 80, 150, ?*

**Q10. Find the next term in the sequence.**

(a) 216
(b) 252
(c) 240
(d) 294

**Answer: (b) 252**

**Logic / Trick:** Rule: $n^3 + n^2$ for consecutive integers n = 1, 2, 3, 4, 5, 6. n=1: 1+1=2; n=2: 8+4=12; n=3: 27+9=36; n=4: 64+16=80; n=5: 125+25=150; n=6: $6^3 + 6^2 = 216 + 36 = 252$.

**[College ITM Mock Test 2 (Q51 - Matchstick)]**

***Directions:*** *Matchstick Geometry: Five identical unit squares are arranged in a row sharing sides, using 16 matchsticks in total.*

**Q11. What is the minimum number of sticks that must be removed so that exactly 3 complete unit squares remain?**

(a) 1
(b) 2
(c) 3
(d) 4

**Answer: (a) 1**

**Logic / Trick:** The stick shared between the 2nd and 3rd squares is an edge of both. Removing that single stick destroys both squares simultaneously, leaving squares 1, 4, and 5 intact (exactly 3 complete unit squares with only 1 removal).

**[College ITM Mock Test 2 (Q52 - Circular Wheel)]**

***Directions:*** *Circular Wheel Logic: Wheel 1: T=6, R=2, B=4, L=3 -> Centre=30. Wheel 2: T=5, R=3, B=8, L=2 -> Centre=46.*

**Q12. Find Centre for Wheel 3: T=7, R=4, B=3, L=6.**

(a) 39
(b) 41
(c) 45
(d) 49

**Answer: (c) 45**

**Logic / Trick:** Rule: Centre = (Top × Bottom) + (Right × Left). Wheel 1: (6×4) + (2×3) = 24 + 6 = 30. Wheel 2: (5×8) + (3×2) = 40 + 6 = 46. Wheel 3: (7×3) + (4×6) = 21 + 24 = 45.

**[College ITM Mock Test 2 (Q53 - Matrix Logic)]**

***Directions:*** *3×3 Numerical Matrix: Row 1: [2, 3, 11]; Row 2: [3, 4, 19]; Row 3: [4, 5, ?].*

**Q13. Find the missing number in the matrix.**

(a) 27
(b) 29
(c) 31
(d) 33

**Answer: (b) 29**

**Logic / Trick:** Rule: Third column = (a × b) + a + b. Row 1: (2×3) + 2 + 3 = 11. Row 2: (3×4) + 3 + 4 = 19. Row 3: (4×5) + 4 + 5 = 20 + 9 = 29.

**[College ITM Mock Test 2 (Q54 - Tetrahedral Sequence)]**

***Directions:*** *Mathematical Sequence: 1, 4, 10, 20, 35, ?*

**Q14. Find the next term in the sequence.**

(a) 52
(b) 56
(c) 60
(d) 49

**Answer: (b) 56**

**Logic / Trick:** Tetrahedral numbers: T(n) = n(n+1)(n+2)/6. T(1)=1, T(2)=4, T(3)=10, T(4)=20, T(5)=35. For n=6: 6×7×8/6 = 56.

**[College ITM Mock Test 3 (Q51 - Matchstick)]**

***Directions:*** *Matchstick Geometry: Three unit squares arranged in an L-shape using 10 matchsticks.*

**Q15. What is the minimum number of sticks to remove so that no complete square remains?**

(a) 1
(b) 2
(c) 3
(d) 4

**Answer: (b) 2**

**Logic / Trick:** Removing the stick shared between top-left and top-right squares breaks both in 1 move. One more removal of any stick of the bottom square breaks the third. Minimum = 2 sticks.

**[College ITM Mock Test 3 (Q52 - Circular Wheel)]**

***Directions:*** *Circular Wheel Logic: Wheel 1: T=5, R=3, B=7, L=2 -> Centre=19. Wheel 2: T=4, R=6, B=9, L=3 -> Centre=63.*

**Q16. Find Centre for Wheel 3: T=8, R=5, B=6, L=4.**

(a) 54
(b) 58
(c) 62
(d) 50

**Answer: (b) 58**

**Logic / Trick:** Rule: Centre = (T+B)(R+L) - (T×B + R×L). Wheel 1: (5+7)(3+2) - (35+6) = 60 - 41 = 19. Wheel 2: (4+9)(6+3) - (36+18) = 117 - 54 = 63. Wheel 3: (8+6)(5+4) - (48+20) = 126 - 68 = 58.

**[College ITM Mock Test 3 (Q53 - Matrix Logic)]**

***Directions:*** *3×3 Matrix: [4, 9, 25], [9, 25, 49], [25, 49, ?].*

**Q17. Find the missing number in the matrix.**

(a) 100
(b) 121
(c) 144
(d) 81

**Answer: (b) 121**

**Logic / Trick:** Entries are squares of consecutive primes along anti-diagonals: $2^2=4$, $3^2=9$, $5^2=25$, $7^2=49$, $11^2=121$. The missing cell is $11^2 = 121$.

**[College ITM Mock Test 3 (Q54 - Product Sequence)]**

***Directions:*** *Mathematical Sequence: 6, 24, 60, 120, 210, ?*

**Q18. Find the next term in the sequence.**

(a) 320
(b) 330
(c) 336
(d) 342

**Answer: (c) 336**

**Logic / Trick:** Rule: Product of 3 consecutive integers: 1×2×3=6, 2×3×4=24, 3×4×5=60, 4×5×6=120, 5×6×7=210. Next: 6×7×8 = 336.

**[College ITM Mock Test 4 (Q51 - Matchstick)]**

***Directions:*** *Matchstick Geometry: A 3×3 grid of unit squares (9 small squares, 14 squares total) built with 24 matchsticks.*

**Q19. What is the minimum number of matchsticks to remove so that no complete square of any size remains?**

(a) 4
(b) 5
(c) 6
(d) 7

**Answer: (c) 6**

**Logic / Trick:** Exhaustive computer proof verifies no set of 5 or fewer sticks destroys all 14 squares (nine 1×1, four 2×2, one 3×3). A set of 6 sticks exists that intersects every square. Minimum = 6.

**[College ITM Mock Test 4 (Q52 - Circular Wheel)]**

***Directions:*** *Circular Wheel Logic: Wheel 1: T=4, R=5, B=6, L=3 -> Centre=39. Wheel 2: T=7, R=2, B=8, L=5 -> Centre=66.*

**Q20. Find Centre for Wheel 3: T=6, R=4, B=9, L=2.**

(a) 58
(b) 62
(c) 66
(d) 54

**Answer: (b) 62**

**Logic / Trick:** Rule: Centre = (T+R)(B+L) - (T×L + R×B). Wheel 1: (4+5)(6+3) - (12+30) = 81 - 42 = 39. Wheel 2: (7+2)(8+5) - (35+16) = 117 - 51 = 66. Wheel 3: (6+4)(9+2) - (12+36) = 110 - 48 = 62.

**[College ITM Mock Test 4 (Q53 - Matrix Logic)]**

***Directions:*** *3×3 Matrix: Row 1: [2, 3, 4]; Row 2: [6, 8, 10]; Row 3: [12, 15, ?].*

**Q21. Find the missing number in the matrix.**

(a) 16
(b) 18
(c) 20
(d) 21

**Answer: (b) 18**

**Logic / Trick:** Rule: cell(r, c) = r × c + r². For Row 3 (r=3): cell(3,1) = 3×1+9 = 12; cell(3,2) = 3×2+9 = 15; cell(3,3) = 3×3+9 = 18.

**[College ITM Mock Test 4 (Q54 - Sum of Squares)]**

***Directions:*** *Mathematical Sequence: 1, 5, 14, 30, 55, ?*

**Q22. Find the next term in the sequence.**

(a) 91
(b) 85
(c) 96
(d) 90

**Answer: (a) 91**

**Logic / Trick:** Pyramidal square numbers (Sum of squares $1^2 + 2^2 + ... + n^2$): 1, 1+4=5, 5+9=14, 14+16=30, 30+25=55. Next: 55 + $6^2$ = 55 + 36 = 91.

**[College ITM Mock Test 5 (Q51 - Matchstick)]**

***Directions:*** *Matchstick Geometry: Five unit squares in a plus/cross pattern using 16 matchsticks in total.*

**Q23. What is the minimum number of sticks to remove so that no complete square remains?**

(a) 3
(b) 4
(c) 5
(d) 6

**Answer: (b) 4**

**Logic / Trick:** Each of the 4 outer arms shares exactly 1 stick with the center square. Removing all 4 central-arm shared sticks destroys the center square and all 4 arms. Minimum = 4.

**[College ITM Mock Test 5 (Q52 - Circular Wheel)]**

***Directions:*** *Circular Wheel Logic: Wheel 1: T=8, R=2, B=6, L=3 -> Centre=41. Wheel 2: T=10, R=4, B=5, L=2 -> Centre=65.*

**Q24. Find Centre for Wheel 3: T=12, R=3, B=9, L=5.**

(a) 85
(b) 81
(c) 90
(d) 78

**Answer: (a) 85**

**Logic / Trick:** Rule: Centre = (T - L)² + (B - R)². Wheel 1: (8-3)² + (6-2)² = 25 + 16 = 41. Wheel 2: (10-2)² + (5-4)² = 64 + 1 = 65. Wheel 3: (12-5)² + (9-3)² = 49 + 36 = 85.

**[College ITM Mock Test 5 (Q53 - Matrix Logic)]**

***Directions:*** *3×3 Matrix: Row 1: [3, 7, 13]; Row 2: [7, 12, 19]; Row 3: [13, 19, ?].*

**Q25. Find the missing number in the matrix.**

(a) 25
(b) 27
(c) 29
(d) 24

**Answer: (b) 27**

**Logic / Trick:** Rule: cell(r, c) = r² + c² + r × c. For r=3, c=3: 3² + 3² + 3×3 = 9 + 9 + 9 = 27.

**[College ITM Mock Test 5 (Q54 - Second Differences)]**

***Directions:*** *Mathematical Sequence: 3, 9, 19, 33, 51, ?*

**Q26. Find the next term in the sequence.**

(a) 73
(b) 69
(c) 77
(d) 71

**Answer: (a) 73**

**Logic / Trick:** Differences: 6, 10, 14, 18 (increasing by +4). Next difference: 18 + 4 = 22. Next term: 51 + 22 = 73.

**[College ITM Mock Test 6 (Q51 - Matchstick)]**

***Directions:*** *Matchstick Geometry: A 2×3 grid of unit squares built from 17 matchsticks (contains six 1×1 and two 2×2 squares).*

**Q27. What is the minimum number of sticks to remove so that no complete square of any size remains?**

(a) 3
(b) 4
(c) 5
(d) 6

**Answer: (b) 4**

**Logic / Trick:** Exhaustive computer proof verifies no set of 3 sticks breaks all 8 squares. A set of 4 sticks suffices. Minimum = 4.

**[College ITM Mock Test 6 (Q52 - Circular Wheel)]**

***Directions:*** *Circular Wheel Logic: Wheel 1: T=4, R=1, B=3, L=2 -> Centre=40. Wheel 2: T=6, R=2, B=2, L=3 -> Centre=39.*

**Q28. Find Centre for Wheel 3: T=5, R=3, B=4, L=1.**

(a) 65
(b) 61
(c) 69
(d) 57

**Answer: (a) 65**

**Logic / Trick:** Rule: Centre = (T + B)² - (R + L)². Wheel 1: (4+3)² - (1+2)² = 49 - 9 = 40. Wheel 2: (6+2)² - (2+3)² = 64 - 25 = 39. Wheel 3: (5+4)² - (3+1)² = 81 - 16 = 65.

**[College ITM Mock Test 6 (Q53 - Matrix Logic)]**

***Directions:*** *3×3 Matrix: Row 1: [5, 7, 18]; Row 2: [6, 11, 25]; Row 3: [8, 9, ?].*

**Q29. Find the missing number in the matrix.**

(a) 55
(b) 73
(c) 64
(d) 47

**Answer: (a) 55**

**Logic / Trick:** Rule: Third column = (First)² - Second. Row 1: 5² - 7 = 25 - 7 = 18. Row 2: 6² - 11 = 36 - 11 = 25. Row 3: 8² - 9 = 64 - 9 = 55.

**[College ITM Mock Test 6 (Q54 - Cubic Series)]**

***Directions:*** *Mathematical Sequence: 2, 10, 30, 68, 130, ?*

**Q30. Find the next term in the sequence.**

(a) 222
(b) 216
(c) 210
(d) 228

**Answer: (a) 222**

**Logic / Trick:** Rule: $n^3 + n$ for n = 1, 2, 3, 4, 5, 6. n=1: 1+1=2; n=2: 8+2=10; n=3: 27+3=30; n=4: 64+4=68; n=5: 125+5=130; n=6: 216+6 = 222.